RESEARCH UPDATE

Our extension: contract-level option relative value

Three current P25 strategies, reported separately with their own controls

Current strategy	Annualized excess Sharpe	Entries
30-day learned sizing + management	1.029	212
Six-month conservative fitted-Q management	0.610	207
Native one-year forecast + learned sizing	0.313	276

2018–29 August 2025; actual contract accounting at $10m. These are separate researched strategies, not maturity-only ablations.

TRADE AND ACCOUNTING

Trade the contracts; measure returns against cash

Actual puts selected by expiry and delta, with a shared cash-and-stock ledger

Item	Contract-level convention
Instrument	Same option type; target absolute delta 0.25
Initial neutrality	Integer contract vega matched within 10%; daily stock delta hedge
Actual costs	$0.03 / option contract + 0.1 vol point per leg; stock execution 1 bp
Financing	Floating cash return; stock borrow spread 0.5%
Metric	Daily cash-excess Sharpe × sqrt(252), sample standard deviation

$10m funded equity. Vega is matched at entry, not continuously. Management pays additional costs when it trades.

SIGNAL BEFORE SIZING

Separate the trading signal from the allocation decision

Broad research universe; each maturity keeps its own economic entry definition

- Short-dated P25 state adds expected variance premium, historical normalization and RV / IV ratios.
- Relative-value, profitability memory and observable pair-risk inputs determine the qualified proposals.
- Compare fixed and learned sizes on common signals, contracts, costs and risk limits.
- The new one-year entry study independently forecasts next-year variance and generates monthly opportunities.

WHAT THE LEARNER DOES

Learn size within a focused strategy

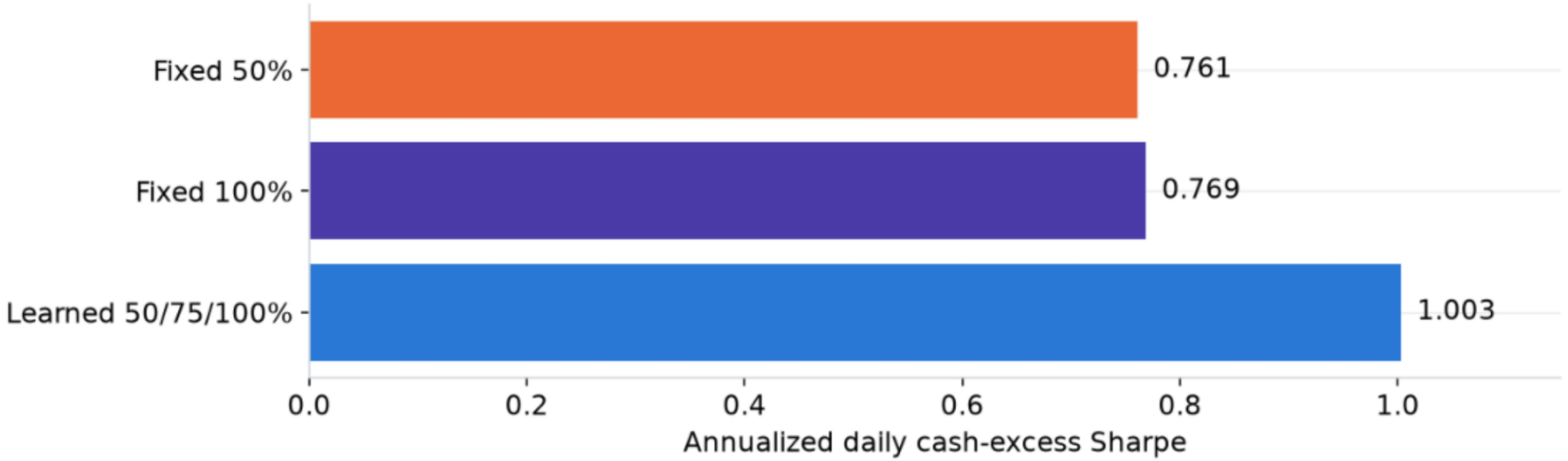
Use precise algorithm labels: contextual allocation and sequential management are distinct

- 30-day sizing: tree states plus direct policy search; actions are 50%, 75% or 100%.
- Six-month management: fitted Q learns weekly hold, half-size or close decisions.
- New one-year sizing: quarterly completed-history fits choose 0%, 25%, 50% or 100%.
- Execution remains subject to quotes, integer contracts and common account risk checks.

30-DAY SIZING

Adaptive sizing improves the fixed-vega controls

Same qualified proposals; 2018–29 August 2025 at $10m



Common proposal set and risk limits. Size changes integer contracts, overlap and which entries fit in the account.

Sizing	Entries	Above-cash dollars
Fixed 50%	263	$248,315
Fixed 100%	133	$272,338
Learned sizing	212	$322,637

MANAGING OPEN POSITIONS

Exit decisions help some maturities more than others

Hold versus learned management, using the controls defined within each study

Maturity	Hold Sharpe	Managed Sharpe	Intervention
30-day	1.003	1.029	Trim / paired exit
90-day	-0.324	0.025	Trim / paired exit
Six-month	0.592	0.610	Conservative fitted Q

- The 90-day locked comparison keeps the original 116 entries.
- Six-month entry is fixed 50%; learned management adds a modest improvement.

INDEPENDENT ONE-YEAR ENTRY

A native long-horizon signal supports a small sizing gain

Forecast next-year variance; do not reuse short-dated entries

One-year policy	Sharpe	Entries
Forecast direction; fixed 50%	0.254	580
Forecast direction; learned size / skip	0.313	276
Same learned skips; fixed 50%	0.273	281
Learn direction and size together	-0.050	316

The sizing-only gain versus matched fixed 50% is +0.040; its 252-session bootstrap 95% interval is approximately [-0.099, +0.203].

SMILE EXPLORATION

The downside-put signal does not transfer uniformly

Earlier smile experiment: separate strike regions, not extra actions in one policy

Instrument / policy	Annualized excess Sharpe
25-delta puts, downside	0.792
75-delta puts, upside strike	0.289
25-delta calls, upside strike	0.018

- The opposite wing was weaker under the tested signals and quote screens.
- The current 30-day, six-month and one-year strategies all target 25-delta puts.

YEAR-BY-YEAR EVIDENCE

Profits and Sharpe vary materially across years

Current managed 30-day, conservative six-month, and native one-year checkpoints

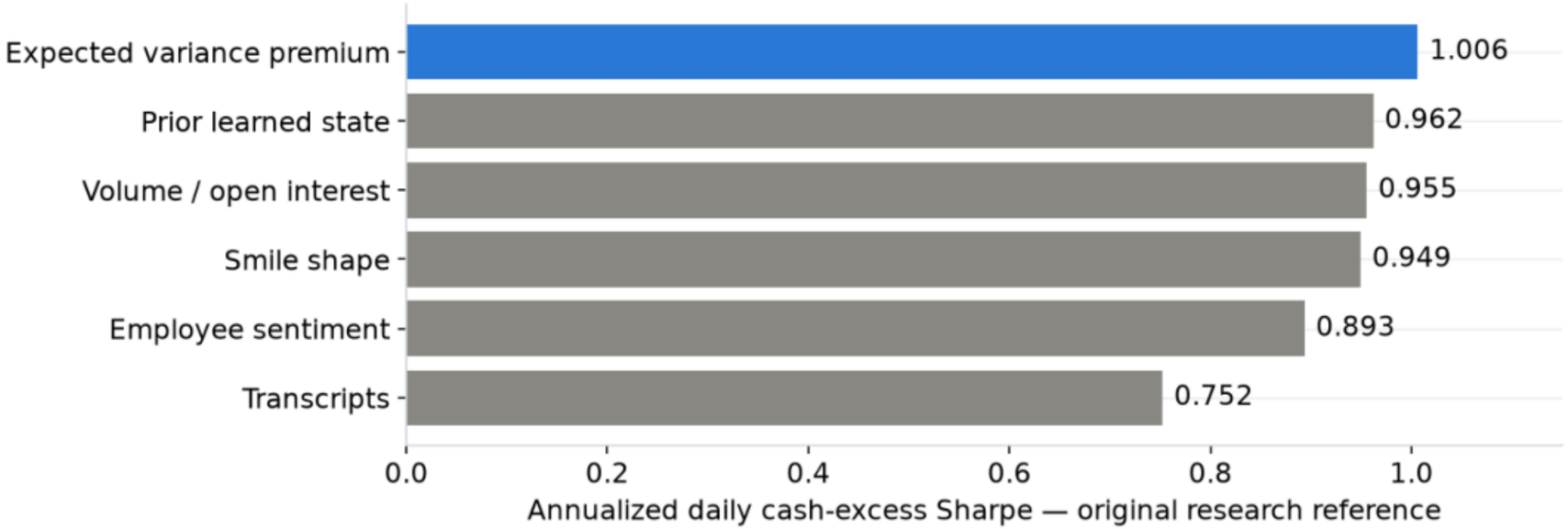
Year	30D return	30D above cash (pp)	30D Sharpe	6M Sharpe	1Y Sharpe
2018	2.14%	+0.06	0.40	-0.35	-0.58
2019	2.20%	-0.08	-0.58	0.06	0.73
2020	1.18%	+0.66	1.58	1.00	0.04
2021	0.15%	+0.01	0.09	-0.02	0.26
2022	3.47%	+1.67	2.46	1.49	0.76
2023	5.29%	+0.43	1.47	0.34	-0.11
2024	5.14%	-0.24	-0.80	0.42	0.61
2025*	3.27%	+0.23	1.24	0.00	0.29

*2025 through 29 August. Percentage returns are calendar-period returns, not annualized. Sharpes are annualized; funded returns include interest on cash.

FEATURE RESEARCH

More features usually failed to improve the retained signal

Same short-dated feature experiment; include successful and unsuccessful additions



Original funding reference and matched short-dated proposals. These are researched configurations, not new held-out tests.

PORTFOLIO ALLOCATION

Past-performance weighting trails simple equal weights

Quarterly fits on prior 24 months; equal weights during the 2018–2019 warmup

Allocation	2018–Aug 2025	2020–Aug 2025
All three: equal weights	1.006	1.185
All three: positive-Sharpe weights	0.923	1.077
All three: covariance optimizer	0.865	1.005
30D + 6M: equal weights	1.029	1.243
30D + 6M: covariance optimizer	0.990	1.187

Annualized excess Sharpes. The retrospective full-period optimized blend reaches 1.107; it is a hindsight benchmark, not the historical allocation result.

INTERPRETATION

Adaptive decisions help within specific strategies

Separate economic signal, learned sizing, sequential management, and portfolio allocation

- Strongest evidence: P25 short-dated sizing improves common fixed-vega controls.
- Smaller gains: six-month management and native one-year sizing; uncertainty remains broad.
- Limits: weak opposite-wing results, failed direction learning, and weaker historical portfolio optimization.
- Next test: freeze the design and evaluate new dates with prospective pair formation.

REPRODUCIBILITY

Keep the original study and the extension together

Current metrics linked to saved reports and the updated research slides

- Ten-page paper · Current options research (PDF)
- Fixed versus learned sizing · Short / 90-day management
- Six-month management · Native one-year entry and sizing
- Past-only allocation results · Presentation update audit